Deploying a MERN Application on AWS using Terraform and Ansible

git clone https://github.com/Adish786/TravelMemory

ec2.tf

```
resource "aws_instance" "adish_web" {
  ami                         = var.ami_id
  instance_type               = var.instance_type
  subnet_id                   = aws_subnet.public_subnet.id
  associate_public_ip_address = true
  vpc_security_group_ids      = [aws_security_group.adish-web_sg.id]
  iam_instance_profile        = aws_iam_instance_profile.ec2_profile.name
  key_name                    = aws_key_pair.mern_keypair.key_name

  tags = {
    Name = var.tags_name_web
  }
}

resource "aws_instance" "adish_db" {
  ami                    = var.ami_id
  instance_type          = var.instance_type
  subnet_id              = aws_subnet.private_subnet.id
  vpc_security_group_ids = [aws_security_group.adish-db_sg.id]
  iam_instance_profile = aws_iam_instance_profile.ec2_profile.name
  key_name               = aws_key_pair.mern_keypair.key_name

  tags = {
    Name = var.tags_name_db
  }
}
```

iam.tf

```
############################################
# IAM Role for EC2 Instances
```

```
############################################

resource "aws_iam_role" "ec2_role" {
  name = "mern-ec2-role"

  assume_role_policy = jsonencode(
    {
      "Version" : "2012-10-17",
      "Statement" : [
        {
          "Effect" : "Allow",
          "Principal" : {
            "Service" : "ec2.amazonaws.com"
          },
          "Action" : "sts:AssumeRole"
        }
      ]
    }
  )
}

############################################
# Attach AWS Managed Policies
############################################

# Allows EC2 to write logs & metrics to CloudWatch
resource "aws_iam_role_policy_attachment" "cloudwatch" {
  role       = aws_iam_role.ec2_role.name
  policy_arn = "arn:aws:iam::aws:policy/CloudWatchAgentServerPolicy"
}

# Enables AWS SSM (Session Manager, Run Command)
resource "aws_iam_role_policy_attachment" "ssm" {
  role       = aws_iam_role.ec2_role.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore"
}

# Full EC2 Access
resource "aws_iam_role_policy_attachment" "ec2_full_access" {
  role       = aws_iam_role.ec2_role.name
```

```
  policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
}

#Administrator Access (for demo purposes - not recommended for production)
resource "aws_iam_role_policy_attachment" "admin_access" {
  role       = aws_iam_role.ec2_role.name
  policy_arn = "arn:aws:iam::aws:policy/AdministratorAccess"
}

############################################
# Instance Profile (Required for EC2)
############################################

resource "aws_iam_instance_profile" "ec2_profile" {
  name = "mern-ec2-instance-profile"
  role = aws_iam_role.ec2_role.name
}
```

keypair.tf

```
# Generate SSH key pair
resource "tls_private_key" "mern_key" {
  algorithm = "RSA"
  rsa_bits  = 2048
}

# Create AWS key pair
resource "aws_key_pair" "mern_keypair" {
  key_name   = var.key_name
  public_key = tls_private_key.mern_key.public_key_openssh
}

# Save private key to local file
resource "local_file" "private_key" {
  filename        = "${path.module}/MERNAppAdishKeyPair.pem"
  content         = tls_private_key.mern_key.private_key_pem
  file_permission = "0400"
}
```

provider.tf

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "6.26.0"
    }
  }
}

# Configure the AWS Provider
provider "aws" {
  region  = "eu-west-2"
  profile = "default"
}
```

outputs.tf

```
output "web_public_ip" {
  value = aws_instance.adish_web.public_ip
}
```

security.tf

```
resource "aws_security_group" "adish-web_sg" {
  vpc_id = aws_vpc.adish-mern_vpc.id

  ingress {
    description = "SSH from EC2 Instance Connect"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
  //  cidr_blocks = ["0.0.0.0/0"]
    cidr_blocks = ["18.206.107.24/29"] # EC2 Instance Connect IP range for eu-west-2
```

```
  }
ingress {
  description = "Temporary SSH"
  from_port   = 22
  to_port     = 22
  protocol    = "tcp"
  cidr_blocks = ["0.0.0.0/0"]
}

  ingress {
    description = "SSH from my IP"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [var.adish-my_ip]
  }

  ingress {
    from_port   = 3000
    to_port     = 3000
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}

resource "aws_security_group" "adish-db_sg" {
  vpc_id = aws_vpc.adish-mern_vpc.id

  ingress {
    from_port       = 27017
    to_port         = 27017
    protocol        = "tcp"
    security_groups = [aws_security_group.adish-web_sg.id]
```

```
  }
}
```

variables.tf

```
variable "adis-vpc_cidr" {
  default = "10.0.0.0/16"
}

variable "public_subnet_cidr" {
  default = "10.0.1.0/24"
}

variable "private_subnet_cidr" {
  default = "10.0.2.0/24"
}

variable "adish-my_ip" {
  description = "My public IP for SSH access"
  default     = "203.0.113.25/32"
}

variable "ami_id" {
  type    = string
  default = "ami-018ff7ece22bf96db"
}

variable "instance_type" {
  type    = string
  default = "t3.medium"
}

variable "key_name" {
  type    = string
  default = "MERNAppAdishKeyPair"
}
```

```
variable "tags_name_web" {
  type    = string
  default = "adish-mern-web-server"
}

variable "tags_name_db" {
  type    = string
  default = "adish-mern-db-server"
}
```

[vpc.tf](vpc.tf)

```
resource "aws_vpc" "adish-mern_vpc" {
  cidr_block           = var.adis-vpc_cidr
  enable_dns_support   = true
  enable_dns_hostnames = true
  tags                 = { Name = "adish-mern-vpc" }
}

//internet Gateway
resource "aws_internet_gateway" "adish-igw" {
  vpc_id = aws_vpc.adish-mern_vpc.id
}

//Sebnets
resource "aws_subnet" "public_subnet" {
  vpc_id                  = aws_vpc.adish-mern_vpc.id
  cidr_block              = var.public_subnet_cidr
  map_public_ip_on_launch = true
}

resource "aws_subnet" "private_subnet" {
  vpc_id     = aws_vpc.adish-mern_vpc.id
  cidr_block = var.private_subnet_cidr
}

//Net Gateway
resource "aws_eip" "nat_eip" {
```

```
  domain = "vpc"
}

resource "aws_nat_gateway" "nat" {
  allocation_id = aws_eip.nat_eip.id
  subnet_id     = aws_subnet.public_subnet.id
  depends_on = [aws_internet_gateway.adish-igw]
}

// Route Tables
resource "aws_route_table" "adish-public-rt" {
  vpc_id = aws_vpc.adish-mern_vpc.id
  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.adish-igw.id
  }
}

resource "aws_route_table" "adish-private-rt" {
  vpc_id = aws_vpc.adish-mern_vpc.id
  route {
    cidr_block     = "0.0.0.0/0"
    nat_gateway_id = aws_nat_gateway.nat.id
  }
}

// Route Table Associations
resource "aws_route_table_association" "public_subnet_assoc" {
  subnet_id      = aws_subnet.public_subnet.id
  route_table_id = aws_route_table.adish-public-rt.id
}

resource "aws_route_table_association" "private_subnet_assoc" {
  subnet_id      = aws_subnet.private_subnet.id
  route_table_id = aws_route_table.adish-private-rt.id
}
```

terraform init

terraform validate

terraform fmt

terraform plan

terraform apply

Install Ansible

sudo apt update
sudo apt upgrade -y
sudo apt install ansible -y
ansible --version

Create Ansible Inventory File

Inside your project folder create:

`inventory.ini`

```
[web]
localhost ansible_connection=local

[db]
10.0.2.45

[all:vars]
ansible_user=ubuntu
ansible_ssh_private_key_file=./MERNAppAdishKeyPair.pem
ansible_python_interpreter=/usr/bin/python3



#[web]
#10.0.1.78 ansible_user=ubuntu
ansible_ssh_private_key_file=~/MERNAppAdishKeyPair.pem

#[db]
```

```
#10.0.2.45 ansible_user=ubuntu
ansible_ssh_private_key_file=~/MERNAppAdishKeyPair.pem
```

inventory.ini

```
[web]
10.0.1.78 ansible_user=ubuntu
ansible_ssh_private_key_file=/home/ubuntu/MERNAppAdishKeyPair.pem
[db]
10.0.2.72 ansible_user=ubuntu
ansible_ssh_private_key_file=/home/ubuntu/MERNAppAdishKeyPair.pem
```

scp -i MERNAppAdishKeyPair.pem MERNAppAdishKeyPair.pem
ubuntu@<WEB_PUBLIC_IP>:/home/ubuntu/

chmod 400 /home/ubuntu/MERNAppAdishKeyPair.pem

Test SSH With Key

ssh -i MERNAppAdishKeyPair.pem ubuntu@10.0.2.154

ansible all -i inventory.ini -m ping

```
ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$ ansible all -i inventory.ini -m ping
10.0.1.78 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
10.0.2.72 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$
```

```
ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$ cat inventory.ini
#[web]
#localhost ansible_connection=local

#[db]
#10.0.2.45

#[all:vars]
#ansible_user=ubuntu
#ansible_ssh_private_key_file=./MERNAppAdishKeyPair.pem
#ansible_python_interpreter=/usr/bin/python3


[web]
10.0.1.78 ansible_user=ubuntu ansible_ssh_private_key_file=/home/ubuntu/MERNAppAdishKeyPair.pem

[db]
10.0.2.72 ansible_user=ubuntu ansible_ssh_private_key_file=/home/ubuntu/MERNAppAdishKeyPair.pem

ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$
```

STEP 2 — Setup Database Server

ansible-playbook -i inventory.ini db.yml

```
ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$ ansible-playbook -i inventory.ini db.yml

PLAY [Setup MongoDB Server] ************************************************************************************************

TASK [Gathering Facts] *****************************************************************************************************
ok: [10.0.2.72]

TASK [Install required packages] *******************************************************************************************
ok: [10.0.2.72]

TASK [Add MongoDB GPG key] *************************************************************************************************
changed: [10.0.2.72]

TASK [Add MongoDB repository] **********************************************************************************************
changed: [10.0.2.72]

TASK [Update apt cache] ****************************************************************************************************
changed: [10.0.2.72]

TASK [Install MongoDB] *****************************************************************************************************
changed: [10.0.2.72]

TASK [Start and enable MongoDB] ********************************************************************************************
changed: [10.0.2.72]

PLAY RECAP *****************************************************************************************************************
10.0.2.72                  : ok=7    changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$
```

After success, verify MongoDB:

```
ansible db -i inventory.ini -a "sudo systemctl status mongodb"
```

```
ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$ ansible db -i inventory.ini -a "sudo systemctl status mongod"
10.0.2.72 | CHANGED | rc=0 >>
● mongod.service - MongoDB Database Server
     Loaded: loaded (/usr/lib/systemd/system/mongod.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-02-16 10:29:00 UTC; 1min 6s ago
       Docs: https://docs.mongodb.org/manual
   Main PID: 3178 (mongod)
     Memory: 68.5M (peak: 68.7M)
        CPU: 1.180s
     CGroup: /system.slice/mongod.service
             └─3178 /usr/bin/mongod --config /etc/mongod.conf

Feb 16 10:29:00 ip-10-0-2-72 systemd[1]: Started mongod.service - MongoDB Database Server.
Feb 16 10:29:00 ip-10-0-2-72 mongod[3178]: {"t":{"$date":"2026-02-16T10:29:00.638Z"},"s":"I",  "c":"CONTROL",  "id":7484500, "ctx":"-","m
sg":"Environment variable MONGODB_CONFIG_OVERRIDE_NOFORK == 1, overriding \"processManagement.fork\" to false"}
ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$
```

STEP 3 — Setup Web Server

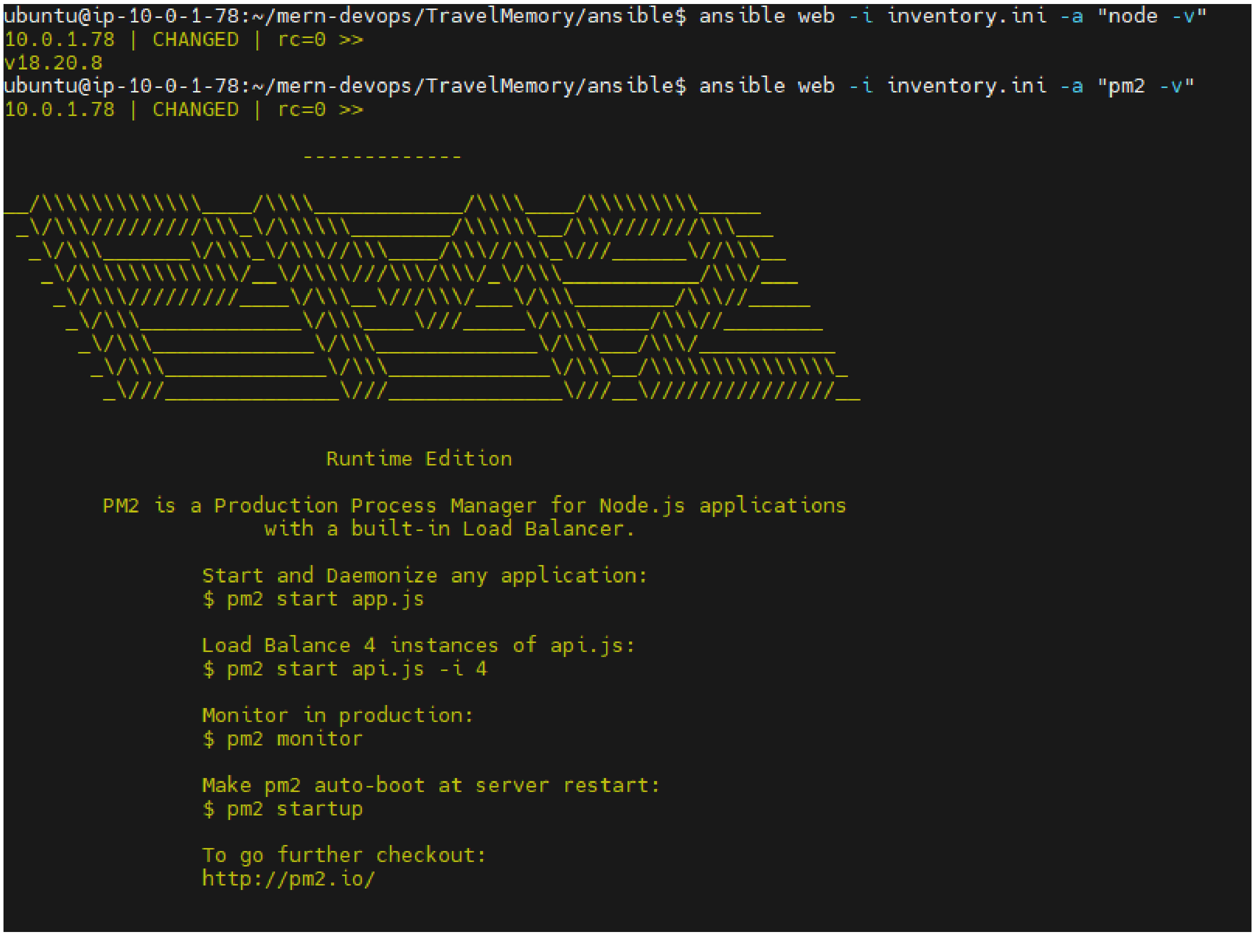
Run:

```
ansible-playbook -i inventory.ini web.yml
```

```
ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$ ansible-playbook -i inventory.ini web.yml

PLAY [Setup Web Server] ********************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [10.0.1.78]

TASK [Update apt] **************************************************************************************************
changed: [10.0.1.78]

TASK [Install Node.js] *********************************************************************************************
changed: [10.0.1.78]

TASK [Install PM2 globally] ****************************************************************************************
changed: [10.0.1.78]

TASK [Install Git] *************************************************************************************************
ok: [10.0.1.78]

PLAY RECAP *********************************************************************************************************
10.0.1.78                  : ok=5    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$
```

Verify:

```
ansible web -i inventory.ini -a "node -v"
ansible web -i inventory.ini -a "pm2 -v"
```



STEP 4 — Deploy MERN Application

Now run:

```
ansible-playbook -i inventory.ini deploy.yml
```

```
ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$ ansible-playbook -i inventory.ini deploy.yml

PLAY [Deploy MERN App] ***********************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************
ok: [10.0.1.78]

TASK [Ensure Git is installed] ***************************************************************************************
ok: [10.0.1.78]

TASK [Clone or update repository] ************************************************************************************
ok: [10.0.1.78]

TASK [Install backend dependencies] **********************************************************************************
ok: [10.0.1.78]

TASK [Install frontend dependencies] *********************************************************************************
ok: [10.0.1.78]

TASK [Build frontend] ************************************************************************************************
changed: [10.0.1.78]

TASK [Stop existing PM2 process if running] **************************************************************************
changed: [10.0.1.78]

TASK [Start backend with PM2 using npm] ******************************************************************************
changed: [10.0.1.78]

TASK [Save PM2 process list] *****************************************************************************************
changed: [10.0.1.78]

PLAY RECAP ***********************************************************************************************************
10.0.1.78                  : ok=9    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ip-10-0-1-78:~/mern-devops/TravelMemory/ansible$
```

1. ansible all -m ping
2. ansible-playbook db.yml
3. ansible-playbook web.yml
4. ansible-playbook deploy.yml
5. ansible-playbook security.yml

web.yml

```yaml
- name: Setup Web Server
  hosts: web
  become: yes

  tasks:

    - name: Update apt
      apt:
        update_cache: yes

    - name: Install Node.js
      shell: |
        curl -fsSL https://deb.nodesource.com/setup_18.x | bash -
        apt install -y nodejs
      args:
        executable: /bin/bash
```

```yaml
    - name: Install PM2 globally
      npm:
        name: pm2
        global: yes

    - name: Install Git
      apt:
        name: git
        state: present
```

db.yml

```yaml
- name: Setup MongoDB Server
  hosts: db
  become: yes

  tasks:

    - name: Install required packages
      apt:
        name:
          - gnupg
          - curl
        state: present
        update_cache: yes

    - name: Add MongoDB GPG key
      shell: |
        curl -fsSL https://pgp.mongodb.com/server-6.0.asc | gpg --dearmor
-o /usr/share/keyrings/mongodb-server-6.0.gpg
      args:
        creates: /usr/share/keyrings/mongodb-server-6.0.gpg

    - name: Add MongoDB repository
      copy:
        dest: /etc/apt/sources.list.d/mongodb-org-6.0.list
        content: |
```

```
          deb [ arch=amd64,arm64
signed-by=/usr/share/keyrings/mongodb-server-6.0.gpg ]
https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 multiverse

    - name: Update apt cache
      apt:
        update_cache: yes

    - name: Install MongoDB
      apt:
        name: mongodb-org
        state: present

    - name: Start and enable MongoDB
      service:
        name: mongod
        state: started
        enabled: yes
```

deploy.yml

```
- name: Deploy MERN App
  hosts: web
  become: yes

  vars:
    app_dir: /home/ubuntu/mern

  tasks:

    - name: Ensure Git is installed
      apt:
        name: git
        state: present
        update_cache: yes

    - name: Clone or update repository
      git:
```

```
        repo: https://github.com/Adish786/TravelMemory.git
        dest: "{{ app_dir }}"
        version: main
        force: yes

    - name: Install backend dependencies
      npm:
        path: "{{ app_dir }}/backend"

    - name: Install frontend dependencies
      npm:
        path: "{{ app_dir }}/frontend"

    - name: Build frontend
      command: npm run build
      args:
        chdir: "{{ app_dir }}/frontend"

    - name: Stop existing PM2 process if running
      shell: pm2 delete travelmemory || true

    - name: Start backend with PM2
      command: pm2 start server.js --name travelmemory
      args:
        chdir: "{{ app_dir }}/backend"

    - name: Save PM2 process list
      command: pm2 save
```

Create .env in THIS Folder

/home/ubuntu/mern/TravelMemory/backend

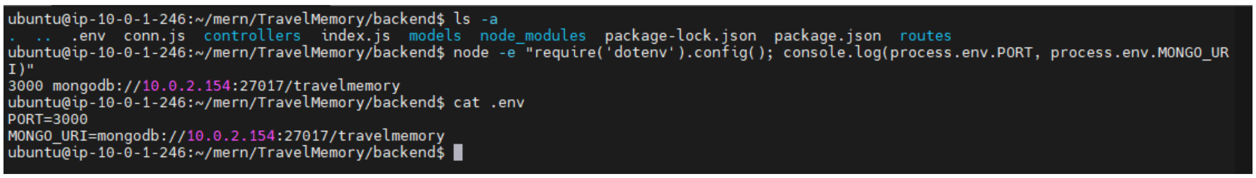
sudo nano .env

Add

PORT=3000
MONGO_URI=mongodb://10.0.2.154:27017/travelmemory

Test Environment Variables

node -e "require('dotenv').config(); console.log(process.env.PORT, process.env.MONGO_URI)"

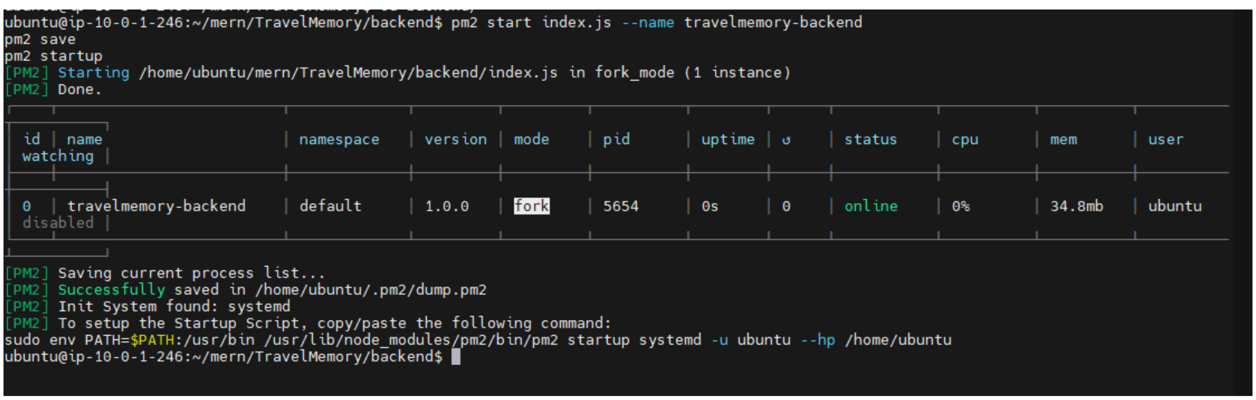


```
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$ ls -a
.  ..  .env  conn.js  controllers  index.js  models  node_modules  package-lock.json  package.json  routes
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$ node -e "require('dotenv').config(); console.log(process.env.PORT, process.env.MONGO_UR
I)"
3000 mongodb://10.0.2.154:27017/travelmemory
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$ cat .env
PORT=3000
MONGO_URI=mongodb://10.0.2.154:27017/travelmemory
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$
```

pm2 start index.js --name travelmemory-backend

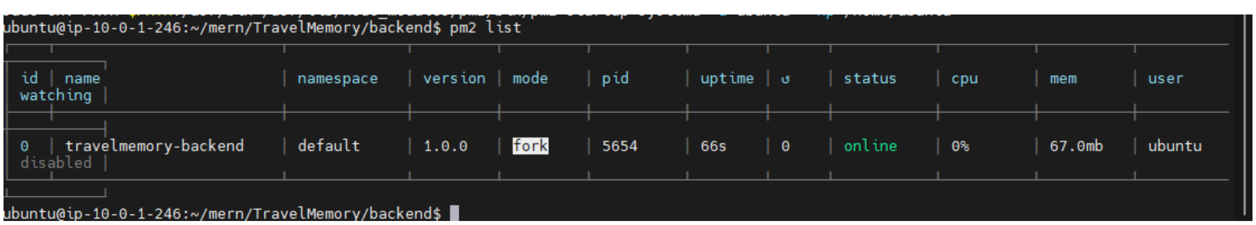
pm2 save

pm2 startup



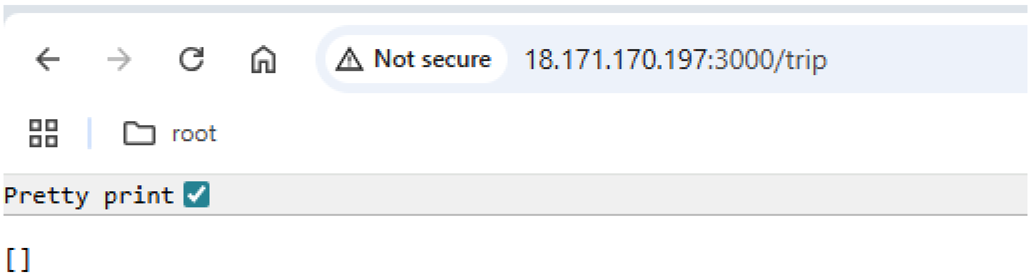
```
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$ pm2 start index.js --name travelmemory-backend
pm2 save
pm2 startup
[PM2] Starting /home/ubuntu/mern/TravelMemory/backend/index.js in fork_mode (1 instance)
[PM2] Done.
│ id │ name                  │ namespace │ version │ mode │ pid  │ uptime │ ↺ │ status │ cpu │ mem    │ user   │
│ watching │
│ 0  │ travelmemory-backend  │ default   │ 1.0.0   │ fork │ 5654 │ 0s     │ 0 │ online │ 0%  │ 34.8mb │ ubuntu │
│ disabled │
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u ubuntu --hp /home/ubuntu
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$
```

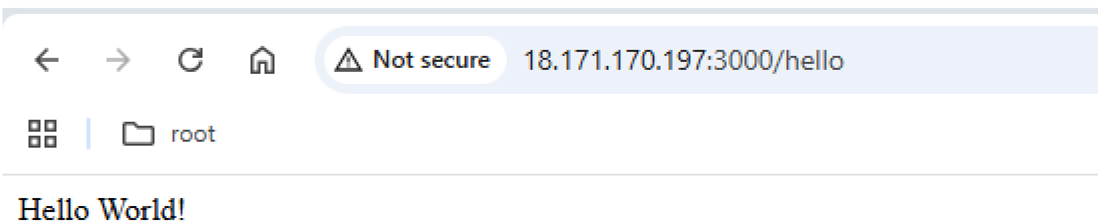
pm2 list



```
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$ pm2 list
│ id │ name                  │ namespace │ version │ mode │ pid  │ uptime │ ↺ │ status │ cpu │ mem    │ user   │
│ watching │
│ 0  │ travelmemory-backend  │ default   │ 1.0.0   │ fork │ 5654 │ 66s    │ 0 │ online │ 0%  │ 67.0mb │ ubuntu │
│ disabled │
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$
```

```
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$ curl http://10.0.1.246:3000/trip
[]ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$ curl http://10.0.1.246:3000/hello
Hello World!ubuntu@ip-10-0-1-246:~/mern/TravelMemory/backend$
```





security.yml

```yaml
- name: Harden EC2 Security
  hosts: all
  become: yes

  vars:
    ssh_port: 22
    app_port: 3000
    mongo_port: 27017

  tasks:

    - name: Disable root SSH login
      lineinfile:
```

```yaml
      path: /etc/ssh/sshd_config
      regexp: '^PermitRootLogin'
      line: 'PermitRootLogin no'

  - name: Disable password authentication
    lineinfile:
      path: /etc/ssh/sshd_config
      regexp: '^PasswordAuthentication'
      line: 'PasswordAuthentication no'

  - name: Restart SSH
    service:
      name: ssh
      state: restarted

  - name: Install UFW firewall
    apt:
      name: ufw
      state: present
      update_cache: yes

  - name: Allow SSH
    ufw:
      rule: allow
      port: "{{ ssh_port }}"
      proto: tcp

  - name: Allow Application Port
    ufw:
      rule: allow
      port: "{{ app_port }}"
      proto: tcp

  - name: Enable UFW
    ufw:
      state: enabled
      policy: deny
```

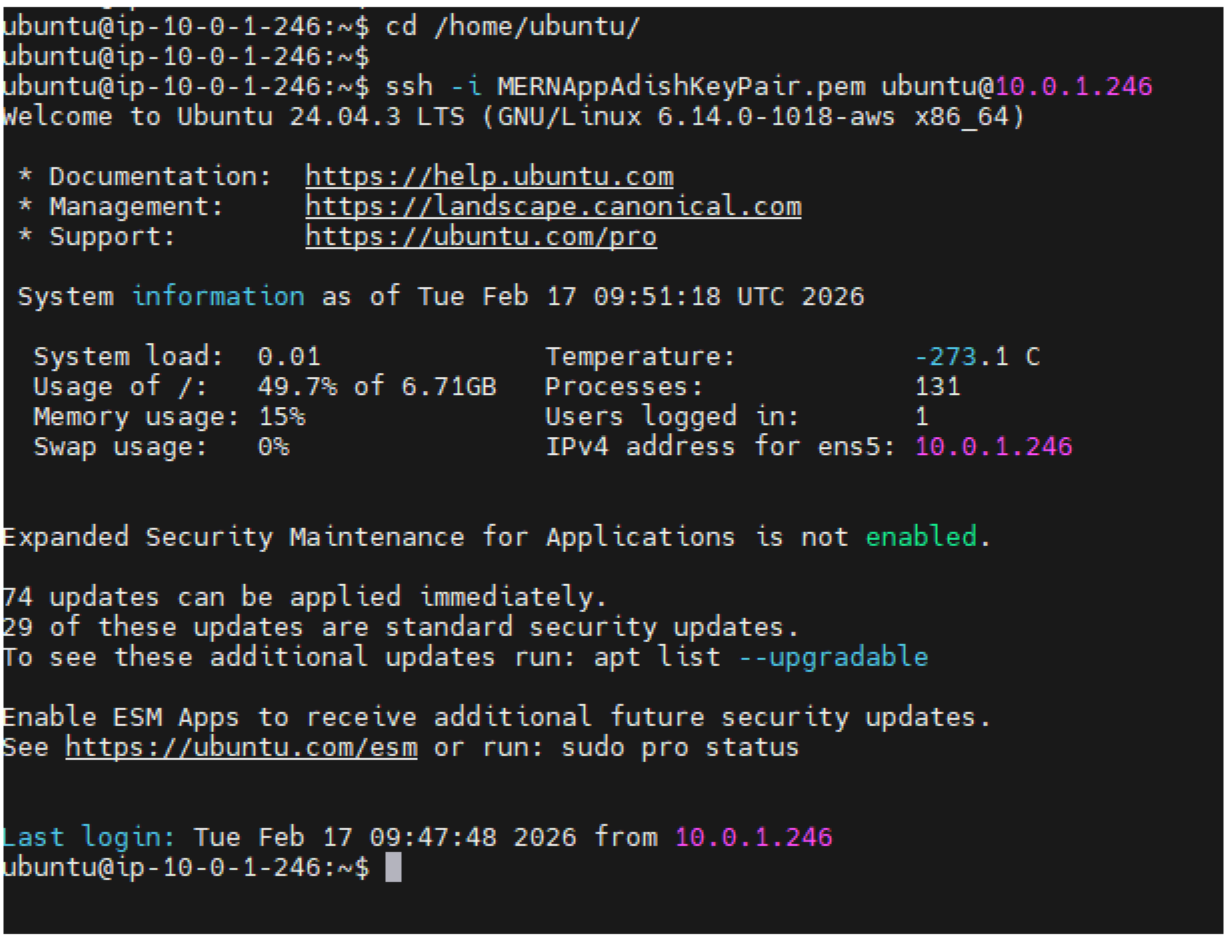
Note :-

- name: Disable root SSH login
- name: Disable password authentication
- name: Enable UFW (deny by default)

Before running this command

ansible-playbook -i inventory.ini security.yml

Fast Test The command
ssh -i yourkey.pem ubuntu@server-ip



```
ubuntu@ip-10-0-1-246:~/mern/TravelMemory/ansible$ ansible-playbook -i inventory.ini security.yml

PLAY [Harden EC2 Security] ****************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************
ok: [10.0.1.246]
ok: [10.0.2.154]

TASK [Disable root SSH login] *************************************************************************************************
ok: [10.0.2.154]
ok: [10.0.1.246]

TASK [Disable password authentication] ****************************************************************************************
ok: [10.0.2.154]
ok: [10.0.1.246]

TASK [Restart SSH] ************************************************************************************************************
changed: [10.0.2.154]
changed: [10.0.1.246]

TASK [Install UFW firewall] ***************************************************************************************************
ok: [10.0.2.154]
ok: [10.0.1.246]

TASK [Allow SSH] **************************************************************************************************************
ok: [10.0.2.154]
ok: [10.0.1.246]

TASK [Allow Application Port] *************************************************************************************************
ok: [10.0.2.154]
ok: [10.0.1.246]

TASK [Enable UFW] *************************************************************************************************************
ok: [10.0.2.154]
ok: [10.0.1.246]

PLAY RECAP ********************************************************************************************************************
10.0.1.246                 : ok=8    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
10.0.2.154                 : ok=8    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ip-10-0-1-246:~/mern/TravelMemory/ansible$
```